

**Mansoura University
Faculty of Computers
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Department of
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Agile Methods

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Lecture 7

Requirement Modeling In Practical (Fact-finding)

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Fact-Finding

Fact Finding for Enterprises

- Zachman Framework
- TOGAF Framework

Interviews

Other fact-finding Techniques

- Document Review
- Observations
- Questionnaires
- Sampling
- Research

Documentation





Requirements Modeling in Practical

Did Henry Ford Say **Faster Horses**? == Pain Point

Product designers must probe deep underlying pain points of customers and come up with the most optimal solutions for a common pain point.

Fun Facts



I Think MR Bean's pain points are fun facts

Requirements Modeling in Practical

We keep hearing "**customer is the king**", listen to him. While building products, we always tend to keep hearing to the customer.

It's very important to hear the customers all the time.

Just hearing the customer and doing what they ask, is a very bad way of **solutioning** and serving the customer.

However, it's even more important to understand the real "**pain point**" of the customer.

What business functions are supported by the current system?

What strategic objectives and business requirements must be supported by the new system?

What are the benefits and TCO of the proposed system?

What transactions will the system process?

What information do users and managers need from the system?

Must the new system interface with legacy systems?

What procedures could be eliminated by business process reengineering?

What security issues exist?

What risks are acceptable?

What budget and timetable constraints will affect system development?

FACT-FINDING FOR ENTERPRISES

Fact-finding: collecting information.

Whether working solo or as a member of a JAD team, during requirements modeling the analyst will use various fact-finding techniques, including

- interviews
- document review
- observation
- surveys and questionnaires
- sampling
- research.
- Typically, this activity begins by asking a series of questions, such as these:

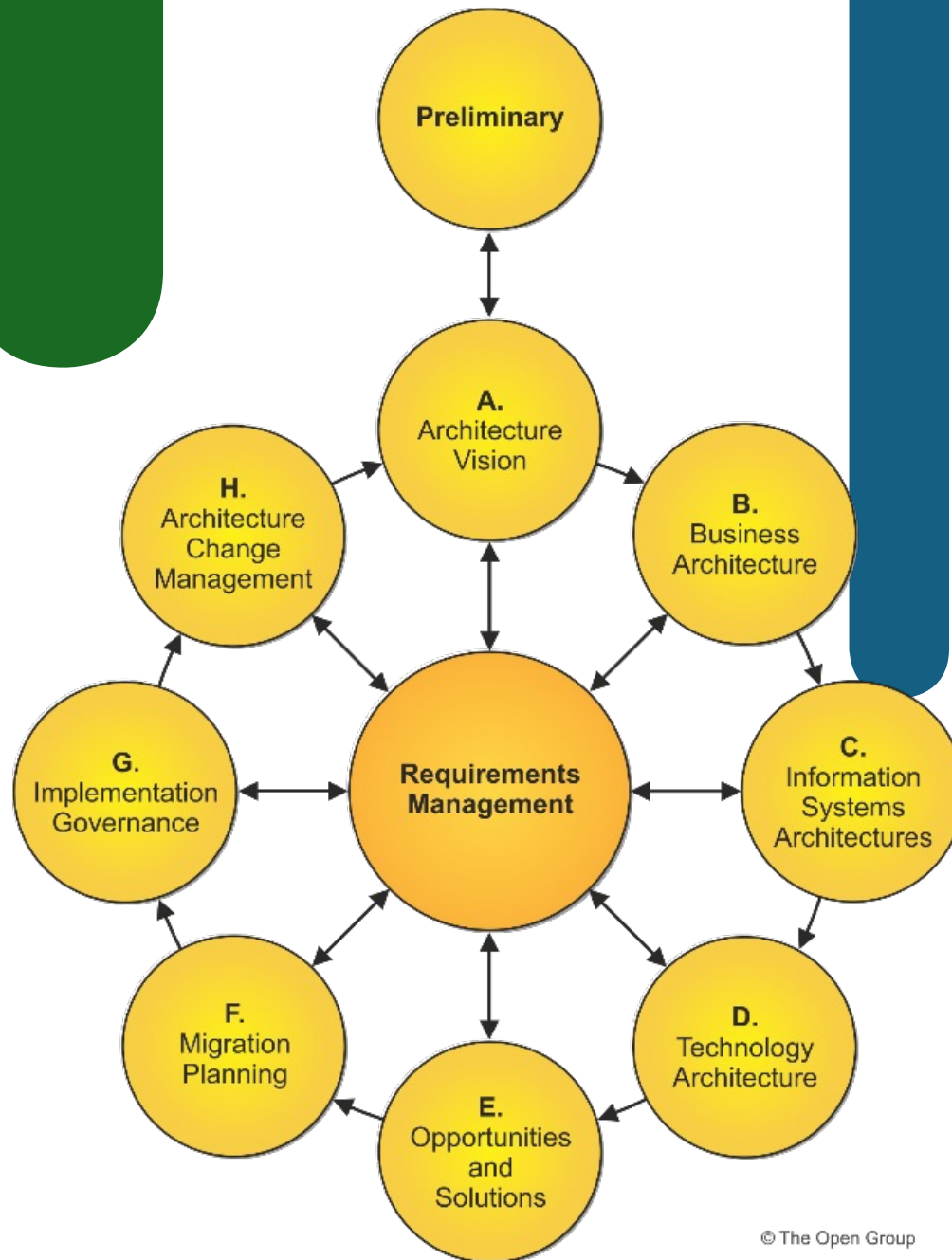
	Why	How	What	Who	Where	When
Contextual	Goal List	Process List	Material List	Organisational Unit & Role List	Geographical Locations List	Event List
Conceptual	Goal Relationship	Process Model	Entity Relationship Model	Organisational Unit & Role Relationship Model	Locations Model	Event Model
Logical	Rules Diagram	Process Diagram	Data Model Diagram	Role Relationship Diagram	Locations Diagram	Event Diagram
Physical	Rules Specification	Process Function Specification	Data Entity Specification	Role Specification	Location Specification	Event Specification
Detailed	Rules Details	Process Details	Data Details	Role Details	Location Details	Event Details

Fact-Finding Plan

To obtain answers to these questions, the analyst develops a fact-finding plan, which can involve another series of questions (who, what, where, when, and how).

OR use a more structured approach, such as the Zachman Framework

The Zachman Framework is a popular approach to modeling enterprise architectures.



TOGAF Framework

TOGAF helps businesses align IT goals with overall business goals, while helping to organize cross-departmental IT efforts.

TOGAF helps businesses define and organize requirements before a project starts, keeping the process moving quickly with few errors.

Created By The Open Group.



“

Your most unhappy customers are
your greatest source of learning

”

~ Bill Gates, Microsoft

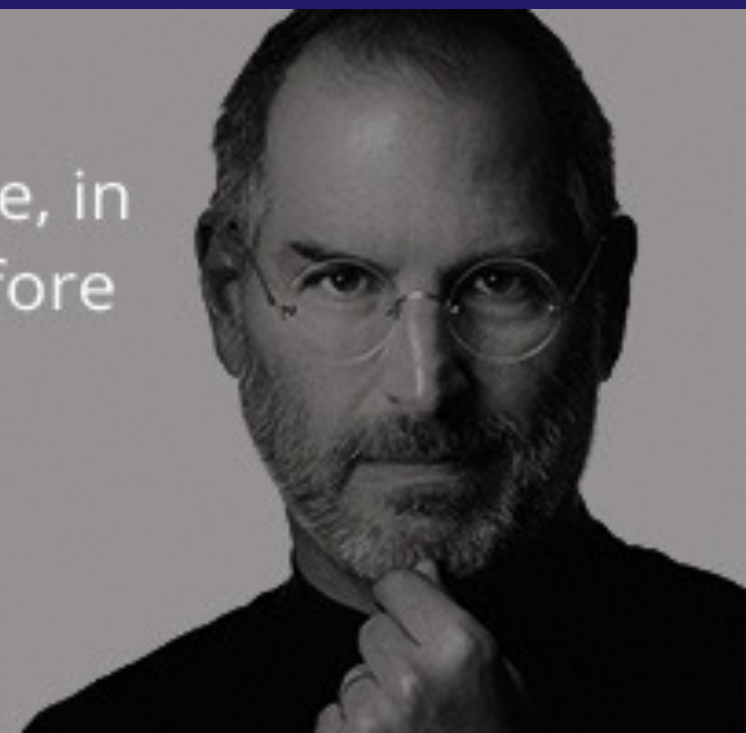
Interviews

“

Get closer than ever to your customers. So close, in
fact, that you tell them what they need well before
they realize it themselves.

”

~ Steve Jobs, Apple





Interviews

Interviewing is an important fact-finding tool during the systems analysis phase.

An interview is a planned meeting during which the analyst obtains **information** from another person.

The skills needed to plan, conduct, document, and evaluate interviews successfully must be understood.

STEP 1: DETERMINE THE PEOPLE TO INTERVIEW:

- The analyst must select the right people to interview and ask them the right questions.
- **Informal structures** usually are based on interpersonal relationships.
- **Group interviews** can save time and provide an opportunity to observe interaction among the participants.
- Group interviews can also present problems.



STEP 2: ESTABLISH OBJECTIVES FOR THE INTERVIEW:

Objectives for the session must be established

- First, the general areas to be discussed should be determined
- then the facts to be gathered should be listed. Soliciting ideas, suggestions, and opinions.

Upper-level managers can provide the big picture to help understand the system as a whole.



Step 3: DEVELOP INTERVIEW QUESTIONS

Creating a standard list of interview questions helps to keep the session on track and avoid unnecessary tangents.

An answer to one question leads to another topic that warrants pursuing.

The interview should consist of several different kinds of questions: open-ended, closed-ended, or questions with a range of responses.

And avoid Leading Questions.



Step 4: PREPARE FOR THE INTERVIEW:

Careful preparation is essential because an interview is an important meeting and not just a casual chat.

A list of topics should be sent to an interviewee several days before the meeting, especially when detailed information is needed, so the person can prepare for the interview and minimize the need for a follow-up meeting.



STEP 5: CONDUCT THE INTERVIEW:

A specific plan for the meeting should be developed.

When conducting an interview, begin with introductions, describe the project, and explain the interview objectives.

Analysts sometimes hear only what they expect to hear. Concentrate on what is said and notice any nonverbal communication that takes place. This process is called engaged listening.



STEP 6: DOCUMENT THE INTERVIEW:

Although taking notes during an interview has both advantages and disadvantages, it should be kept to a minimum.

It is a good idea to write down a few notes to remember key points after the interview, but avoid writing down everything that is said.

Studies have shown that 50% of a conversation is forgotten within 30 minutes. Therefore, use the notes to record the facts immediately so they will not be forgotten.

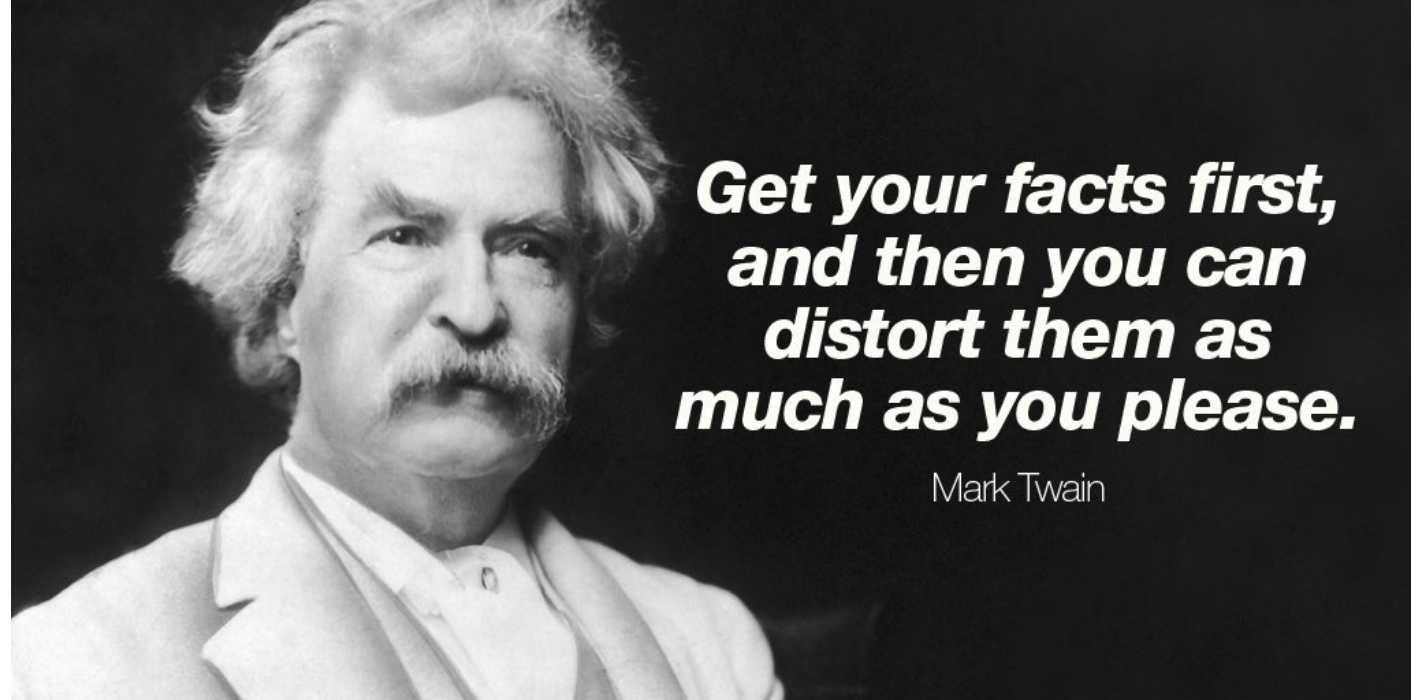


STEP 6: Evaluate THE INTERVIEW:

Try to identify any possible biases.

Interviewee who tries to protect his or her own area or function might give incomplete answers or refrain from volunteering information.

Interviewee with strong opinions about the current or future system might distort the facts.





Other Fact Finding-Techniques

Systems analysts use other fact-finding techniques, including document review, observation, questionnaires and surveys, sampling, and research.



DOCUMENT REVIEW

Document review can help the analyst understand how the current system is supposed to work.

Remember that system documentation sometimes is **out of date**.

Forms can **change** or be **discontinued**, and documented procedures often are modified or eliminated.



OBSERVATIONS

Seeing the system in action provides additional perspective and a better understanding of system procedures.

Personal observation also allows the analyst to verify **statements** made in interviews and determine whether procedures really operate as they are described.

consider a factor called the **Hawthorne Effect**.

The purpose of the study was to determine how various changes in the work environment would affect employee productivity.



JOB ANALYSIS QUESTIONNAIRE

Please complete the following, if applicable:

☐ Request for Job Evaluation	☐ Request for Job Re-Evaluation	
Requested Job Title: <u>Gift Processing Specialist</u>	Requested Pay Grade: <u> </u>	
☐ New Position	☐ Vacant Position	☐ Occupied Position

A. EMPLOYEE DATA (PLEASE PRINT):

Employee Name: Telephone: Date:

Division/College: Department:

Job Cluster: Working Title (if different):

Job Title: Job Code (if applicable):

Supervisor's Name:

Supervisor's Title:

Designee or Division Head/Vice President Name

Designee or Division Head/Vice President Title

FLSA: <u> </u>	Pay Grade: <u> </u>	EEO Code: <u> </u>
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Employment Type: ☐ Regular Full-time ☐ Temporary Full-time
☐ Regular Part-time ☐ Temporary Part-time

Number of hours scheduled to work each week:

How long has employee been in the current position: Years Months

What is the employee's effective hire date with the University:

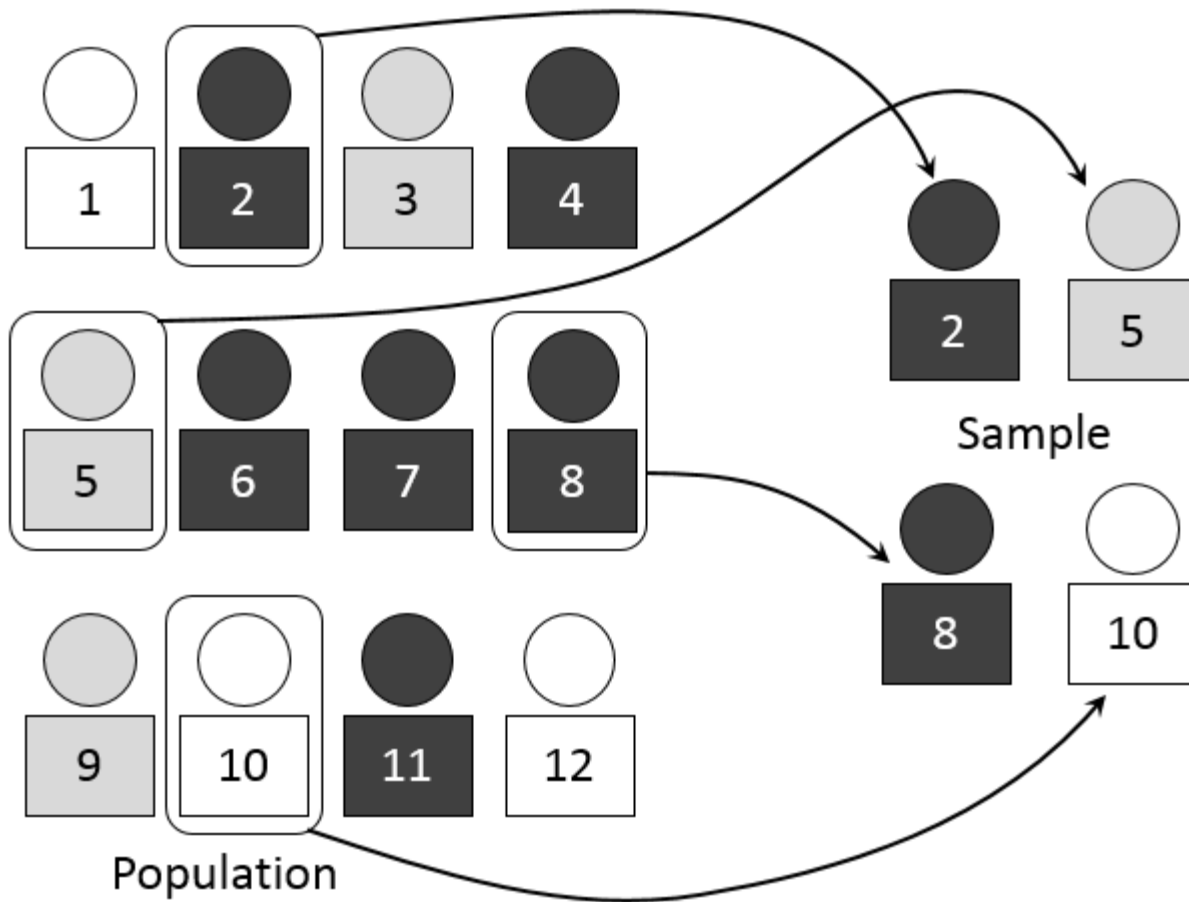
B. PRIMARY PURPOSE OF POSITION

Please briefly summarize the purpose of your position. (Limit summary to 1 to 2 sentences.)

QUESTIONNAIRES AND SURVEYS

A questionnaire, also called a survey, is a document containing a number of standard questions that can be sent to many individuals.

Questionnaires can be used to obtain information about a wide range of topics, including workloads, reports received, volumes of transactions handled, job duties, difficulties, and opinions of how the job could be performed better or more efficiently.



SAMPLING

Sampling techniques include **systematic sampling**, **stratified sampling**, and **random sampling**.

Examples of actual documents should be collected using a process called sampling.

The samples might include records, reports, operational logs, data entry documents, complaint summaries, work requests, and various types of forms.



RESEARCH

Research can include the Internet, IT magazines, and books to obtain background information, technical material, and news about industry trends and developments.

Attending professional meetings, seminars, and discussions with other IT professionals can be very helpful in problem solving.



Documentation

Keeping accurate records of interviews, facts, ideas, and observations is essential to successful systems development.



SOFTWARE Documentation TOOLS

Many software programs are available to help record and document information.

- **CASE Tools**

- Visual Paradigm

- **Productivity Software**

- Word

- Excel

- **Graphic Modeling Software**

- Microsoft Visio



THANK YOU!

